

Fig. 3: Circuit diagram of the alarm unit

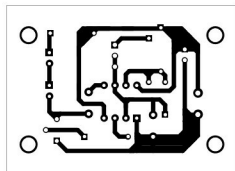


Fig. 4: Actual-size PCB layout of the IR transmitter unit

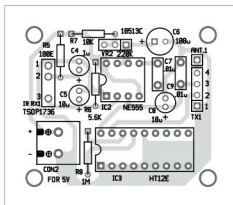


Fig. 7: Component layout of the PCB shown in Fig. 6

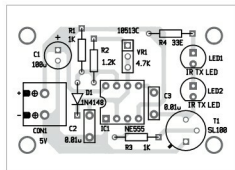


Fig. 5: Component layout of the PCB shown in Fig. 4

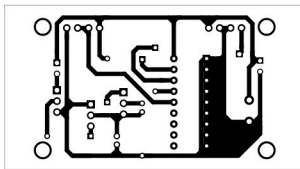


Fig. 8: Actual-size PCB layout of the alarm unit

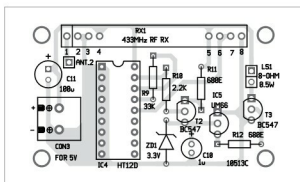


Fig. 9: Component layout of the PCB shown in Fig. 8

Then, output of TSOP1736 sensor goes high. IC2 gets triggered and output pin 3 goes high.

This output signal of around 5V is applied to IC3 and TX1, which transmits the signals to the remote alarm unit. The transmitted signal received by RX1 is fed to pin 14 (DIN) of IC4. Pin 17 of IC4 goes high when a valid RF transmission signal is received.

Transistor T2 (BC547) conducts when pin 17 of IC4 goes high. When T2 conducts, IC5 gets power supply and it produces an alarm sound through loudspeaker LS1.

Construction and testing

An actual-size, single-side PCB layout for the IR transmitter unit is shown in Fig. 4 and its component layout in Fig. 5. Assemble the unit in a suitable enclosure and fix on the frame of the gate.

An actual-size, single-side PCB for the IR receiving unit is shown in Fig. 6 and its component layout in Fig. 7. Assemble the unit in a suitable enclosure and fit at the other side of the gate.

An actual-size, single-side PCB layout for the alarm unit is shown in Fig. 8 and its component layout in Fig. 9. Assemble the unit in a suitable enclosure and keep it inside the house.

All three units work on 5V DC power supply. Use an antenna for transmitter and receiver RF modules; a 17cm single-strand hook-up wire can be used as the antenna. **EFY**

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